

Sriram Hariharan Neelakantan

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PROFESSIONAL EXPERIENCE

L.B. Foster Salient Systems, Data Scientist II | Dublin, OH

Dec 2024 - May 2025

- Built a scalable data pipeline using Python & PySpark on Databricks with Azure Blob Storage to process 3M+ high-speed sensor data, enabling automated wheel filtering, anomaly detection, poly curve analysis, & weight calculation.
- Developed a wheel health metric using double differential filtering and polynomial curve fitting with K-means to filter out 35% of jittery wheels and detect 18% more true electronic anomalies, boosting predictive accuracy.
- Automated ADC-based sensor data validation across 16 circuits per rail enhances electronic signal integrity and strengthens anomaly isolation and wheel health diagnostics by reducing misclassification by 40%.

Norfolk Southern Corp, Machine Learning | USA

Sep 2022 - Dec 2022

- Leveraged K-Nearest Neighbors, PCA, and K-Means within an agile framework to cluster and enhance the spatial analysis of 90k+ rail-road intersections, advancing infrastructure planning efforts
- Streamlined Exploratory Data Analysis with Pandas, NumPy, and Matplotlib to pinpoint over 50 anomalies among 30,000+ data points, achieving a 40% boost in clustering accuracy and optimizing project deliverables
- Implemented Faster R-CNN using PyTorch for railway defect (in coaches, tracks, plates, etc) detection, categorizing over 1000 defects from 7k undercarriage camera images, helping enhance maintenance precision & safety

Accenture, Data Analyst II | India

Mar 2020 - May 2021

- Architected a data pipeline using supervised ML models and Apache Spark for insurance campaign response studies, enhancing conversion rates by 5.5% through customized campaigns
- Improved quarterly risk assessment accuracy by 12% by implementing a gradient boosting model that predicts policyholder default probabilities, using demographic and historical claim data to refine risk mitigation strategies
- Identified key features causing customer early lapse through lapse analysis, reporting findings using Python, SQL, and Tableau to the cross-functional Finance team, by refining pricing strategies

Accenture, Data Analyst I | India

Jan 2019 - Mar 2020

- Refined ARIMA-based sales seasonal forecasting for internal products using 250k records to identify trends, by conducting A/B testing on 12 insights, increasing market base & gross margin by 8%
- Achieved a 15% increase in effective treatment outcomes by combining Link Analysis & XGBoost into the healthcare product efficacy model, reducing false positives by 20%, using Databricks for data processing & tracking
- Deduced key biomarkers linked to cardiovascular health in healthcare product performance assessments by using Python for statistical analysis and Tableau for data visualizations, with AWS S3 and Databricks for data storage

ACADEMIC PROJECTS

Healthcare ChatBot

- Used fine-tuned BERT uncased to create a JSON-based knowledge graph mapping medical entities, enabling graph-based retrieval for over 50 disease profiles and enhancing context-aware medical guidance.
- Curated 100+ medical prompt-completion pairs, fine-tuning FLAN-T5 via Hugging Face Transformers and custom tokenizers, boosting response coherence and medical relevance by ~25%.

Comprehensive Natural Language Question Answering System

- Engineered a BiLSTM-based QA model using DistilBERT embeddings and CrossEntropy loss, optimizing performance through ablation studies on preprocessing and testing architectures up to 10 LSTM layers.
- Demonstrated BERT's superior performance over BiLSTM on SQuAD v1.1 with an average F1 score of 74% vs. 69%, requiring minimal preprocessing while BiLSTM relied on extensive text normalization to perform competitively.

SKILLS

Programming Languages & Frameworks: Python (NumPy, Pandas, Scikit-Learn, PyTorch, Tensor Flow, Keras, OpenCV), R, PySpark, Java (REST APIs), Git, Bitbucket, JIRA, Linux (UNIX)

Databases & Cloud Technologies: SQL, PostgreSQL, MySQL, MongoDB, Apache Spark, AWS EC2, AWS S3, Hadoop, Redshift

Data Science & AI: Predictive Analytics, Machine Learning (Supervised/Unsupervised), Deep Learning, Natural Language Processing (NLP), Computer Vision (CV), Large Language Models (LLM), Probability & Statistics, Hypothesis Testing, Time Series Analysis & Forecasting (ARIMA)

Data Visualization & Tools: Tableau, Excel, Matplotlib, Seaborn, PowerBI, Databricks

EDUCATION

Master of Science, MS in Data Science

Sep 2021 - Dec 2023

Northeastern University, Boston, MA